

Potential of Different Forms of Gravity Energy Storage

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Highlights:

- Compared gravity storage methods holistically by: structure, application, and potential.
- Quantified storage capacity and power output of four solid gravity storage forms.
- Identified storage cycles for various solid gravity energy storage methods.
- Oriented preferred solid gravity storage forms based on practical demands.

Abstract: With the continuous increase in the proportion of renewable energy on the power grid, the stability of the grid is affected, and energy storage technology emerges as a major solution to address such challenges. In comparison to traditional energy storage technologies like batteries and pumped storage, gravity energy storage stands out as an environmentally friendly, cost-effective, and easily implementable energy storage method. This paper conducts a comparative analysis of four primary gravity energy storage forms in terms of technical principles, application practices, and potentials. These forms include Tower Gravity Energy Storage (TGES), Mountain Gravity Energy Storage (MGES), Advanced Rail Energy Storage (ARES), and Shaft Gravity Energy Storage (SGES). The advantages and disadvantages of each technology are analyzed to provide insights for the development of gravity energy storage.

Keywords: renewable energy, gravity energy storage, potential analysis, comparative analysis

Abbreviations

ARES	Advanced Rail Energy Storage
CAES	Compressed Air Energy Storage
CES	Capacitor Energy Storage
FES	Flywheel Energy Storage
GES	Gravity Energy Storage
HES	Hydrogen Energy Storage
IIASA	The Austrian International Research Institute
LCOE	The Levelized Cost of Energy
MGES	Mountain Gravity Energy Storage

PHES	Pumped Storage Hydroelectric Power Generation
PV	Photovoltaic
SGES	Shaft Gravity Energy Storage
TGES	Tower Gravity Energy Storage

24 Nomenclature

D	The height of the mine in SGES system
D'	The available depth of the mine in SGES system
d	The diameter of the suspended heavy object in SGES system
E	Energy storage of the corresponding system
e_h	The head loss in MGES system
e_s	The motor efficiency in MGES system
F	The traction force of the electric motor in ARES system to ensure a constant-speed train
g	Gravitational acceleration
h	Relative height difference
h_s	The height of the suspended object in SGES system
m	The total mass of all the lifting cable cars carrying heavy objects in MGES system
m_A	The total mass and load of the shuttle vehicle in ARES system
m_M	The total mass of the sand or gravel in MGES system
m_S	The mass of suspended heavy matter in SGES system
m_T	The total mass of the bricks in TGES system
P	The output power of the corresponding system
t	Descending time
V	The volume of the suspended heavy object in SGES system
v	The descending or rising speed of the corresponding object
v_{max}	Maximal descending speed
α	Slope
η	Round-trip efficiency
λ	Friction coefficient
μ	Fall or rise efficiency
ρ	Density

25 1. Introduction

26 Due to environmental pollution and the exacerbation of the greenhouse effect
27 caused by fossil fuels, renewable energy sources such as photovoltaic (PV) power,
28 wind power, and hydropower have experienced rapid growth in recent years [1,2].
29 Their share is projected to constitute 26% of the global total energy portfolio by
30 2030 [3]. However, influenced by the natural environment, the power output of
31 renewable energy exhibits intermittency and volatility, posing a threat to the stable
32 operation of power grids [4-6].

Energy storage [7] represents a primary method for mitigating the intermittent impact of renewable energy. By dispatching stored energy to meet demand, a balance between supply and demand can be achieved. This involves storing energy during periods of reduced grid demand and releasing it during periods of increased demand [8]. The integration of energy storage with renewable energy sources can effectively reduce carbon emissions and contribute to achieving carbon neutrality. Analysts from Bloomberg New Energy Finance estimate that by 2040, the global energy storage market (excluding pumped storage) will reach a cumulative capacity of 942,000 MW/2,857,000 MWh, attracting \$ 620 billion in investment [9].

Existing mature energy storage technologies with large-scale applications primarily include pumped storage [10], electrochemical energy storage [11], and Compressed air energy storage (CAES) [12]. The principle of pumped storage involves using electrical energy to drive a pump, transporting water from a lower reservoir to an upper reservoir, and converting it into gravitational potential energy. Switzerland proposed the first pumped storage hydroelectric power generation (PHES) system in 1907 [13]. Pumped storage offers substantial storage capacity, a long storage cycle, low operating costs, and is environmentally friendly. However, it relies on water sources and cannot be utilized in some mountainous areas with limited water availability. Additionally, its construction cost is high, potentially causing environmental damage [14]. Electrochemical energy storage primarily utilizes various types of batteries, such as lead-acid batteries and lithium-ion batteries, to store energy. The objective is to achieve energy storage through the mutual conversion of electric energy and chemical energy within the battery medium. Electrochemical energy storage is convenient and flexible, suitable for short-term energy storage. However, its cost is relatively high, and the electrolyte is toxic, posing a significant environmental impact [15].

CAES involves consuming electrical energy to drive an air compressor, compressing air into a specific storage space for energy storage. When electrical

energy is required, the compressed air is released to drive the turbine and convert it back into electrical energy. CAES provides a fast-instantaneous response speed but incurs high investment costs and is contingent on geographical conditions [16].

Other energy storage technologies with small-scale applications include hydrogen energy storage (HES), flywheel energy storage (FES), and capacitor energy storage (CES), among others. HES involves storing surplus electrical energy by producing hydrogen through the electrolysis of water. The stored hydrogen is then burned to generate electric energy when needed. The current major challenge in HES is the difficulty of storing hydrogen, especially on a large scale. With advancements in hydrogen storage technology, HES is expected to become an indispensable component of the energy storage field [17]. FES consumes electrical energy to drive a flywheel to rotate at a high speed. When electric energy is needed, the flywheel drives a generator to produce electric energy. The principle of FES is simple, with a long lifespan and environmental friendliness. However, a key limitation is the short energy storage time, and the round-trip efficiency decreases over time, making it suitable primarily for short-term energy storage requirements. Additionally, there is a high initial investment cost, and future technological improvements are needed to reduce costs [18]. CES operates similarly to battery energy storage, storing excess electric energy in the capacitor and releasing it when needed. The key difference lies in the fact that CES involves only physical reactions during energy transfer, while battery storage includes chemical reactions exchanging electric and chemical energy. Capacitors offer advantages such as faster charging and discharging but have the drawback of smaller capacity compared to batteries. In the energy storage field, a new electrochemical energy storage method, supercapacitor energy storage, has emerged by combining the advantages of capacitors and batteries. This involves improving power density, extending cycle life, and maintaining fast charging and discharging [19]. The future direction for CES development is to further expand energy storage capacity while retaining the advantage of rapid charging and

91 discharging.

92 In practice, different energy storage technologies are primarily assessed
93 through economic analysis. The most common economic metric for evaluating
94 energy storage projects is the calculation of the levelized cost of energy (LCOE),
95 representing the cost of unit power generation over the entire lifecycle of the
96 projects. Low cost constitutes a significant factor contributing to the optimal
97 development and larger scale of pumped storage. The LCOE is influenced by
98 factors such as the power market, service characteristics, and storage system
99 features. According to the results, PHES achieved a low cost of €120/MWh, while
100 the LCOE costs for CAES and lead-acid battery energy storage [20], utilizing more
101 mature technologies, were €159/MWh and €353/MWh, respectively [21, 22]. After
102 years of development, Ali et al. [23] asserted the selection of a suitable pumped
103 storage area in Australia, predicting the LCOE of the project to range from
104 \$25.46/MWh to \$171.83/MWh.

105 However, the necessity of water sources has always been a bottleneck in the
106 geographical location selection of pumped storage projects. SGES can address this
107 challenge and serve as a substitute or supplement for pumped storage. In a broad
108 sense, gravity energy storage (GES) refers to mechanical technologies that utilize
109 the height drop of energy storage media, such as water or solid, to realize the
110 charging and discharging process of energy storage. Pumped energy storage is also
111 a form of GES. In this paper, SGES refers to a type of energy storage where two
112 energy storage platforms are established, and a unique solid energy storage
113 medium is transported through distinct transportation modes for each energy
114 storage platform. This process completes the mutual conversion between
115 gravitational potential energy and electric energy. Different types of SGES are
116 classified based on various solid energy storage platforms, transportation modes,
117 and solid energy storage media.

118 SGES systems offer flexible site selection compared to pumped storage,
119 higher conversion efficiency, longer energy storage duration, and a lengthier

lifespan than electrochemical energy storage and CAES. Additionally, they have lower costs compared to other energy storage technologies. Their charging and discharging time is measured in hours, with a short reaction time, enabling the quick release of electrical energy for integration into the power grid [24]. Due to the maturity of pumped storage technology, GES has a solid theoretical foundation, and various forms have been developed to address diverse external conditions. For instance, TGES, which lacks specific location requirements, is well-suited for areas abundant in clean energy. MGES and ARES are better equipped to harness the energy storage potential of mountainous regions with varying heights. SGES, on the other hand, can utilize abandoned mines to tap into underground energy storage potential. Since most documents only elaborate on one form of energy storage, this paper will discuss and explain the relevant aspects of SGES based on categories, avoiding a uniform discussion.

Zakeri et al. [21] conducted a calculation of LOCE in SGES utilizing water as the transportation medium and compared it with other energy storage methods. Berrada et al. [22] further computed the costs associated with different components and various construction stages of the system. However, their analyses lacked consideration for the influencing factors specific to the implementation process of the system and lacked verification from actual projects. Furthermore, only a limited number of papers and technology companies have systematically addressed all SGES methods; most tend to focus on a singular energy storage approach.

The development of SGES technologies faces two main challenges: (1) despite research papers showcasing their advantages compared to other energy storage methods and the construction of some demonstration projects, large-scale gravity energy storage projects are currently scarce, and the theoretical data for gravity energy storage remains less convincing; (2) there is a relative scarcity of research on the specific economic analysis of gravity energy storage, especially a lack of comprehensive economic analyses, and insufficient mutual verification between practice and theory.

This paper compares and analyzes four mainstream SGES technologies (TGES, MGES, ARES, and SGES) from perspective of integral structure, application practice and potential. The energy storage capacity as well as the power output are calculated, and the corresponding improved models and application practices are summarized. Furthermore, preferred applicable areas for different forms of solid media gravity energy storage are identified, along with an illustration of the associated challenges.

2. Tower Gravity Energy Storage (TGES)

2.1 Technical introduction and energy storage/release calculation

TGES was first proposed by the Energy Vault company, which utilizes a crane to stack concrete blocks into a tower. Energy is stored and released by lifting and dropping the concrete blocks, as illustrated in Figure 1.

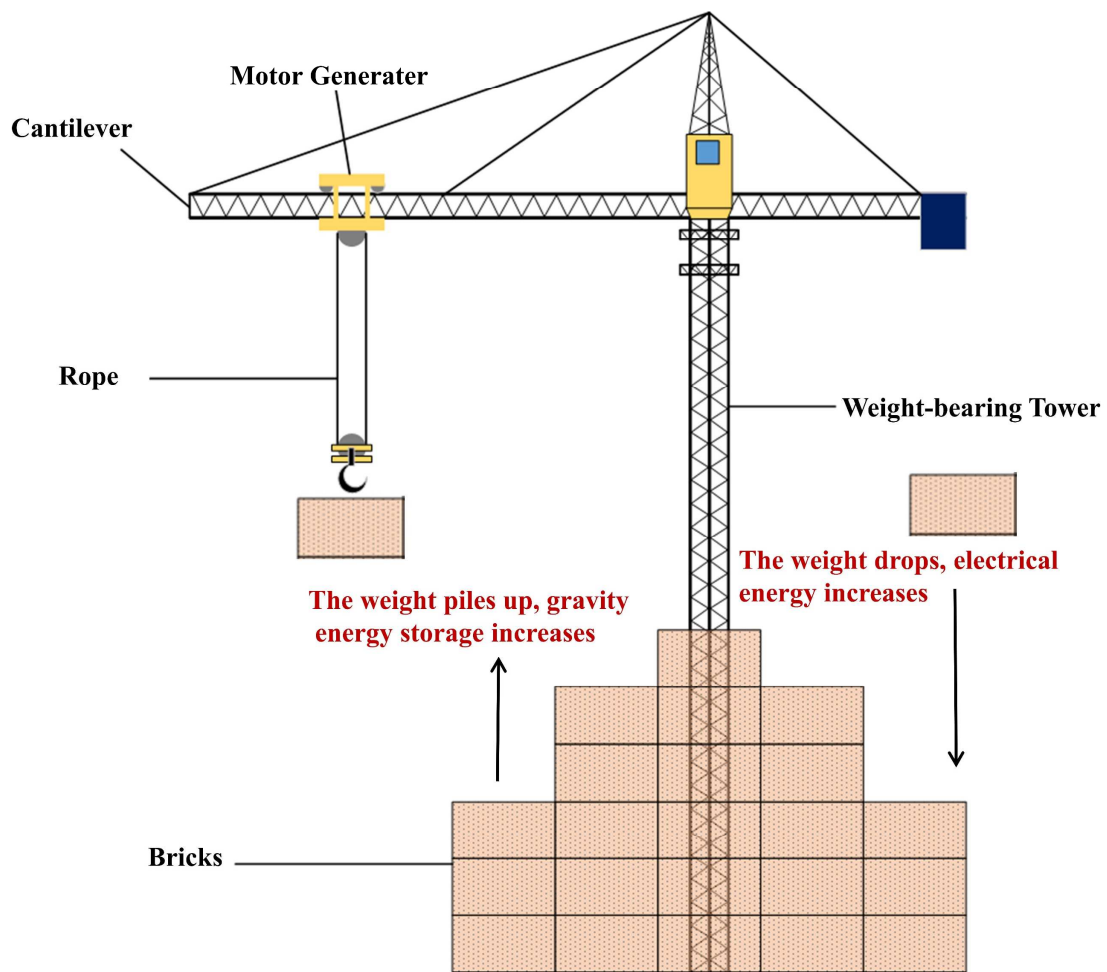


Fig.1 Schematic diagram of TGES device

The energy storage capacity (E) of a TGES device in Fig.1 is calculated by (A1) [25]. It can be observed that the total capacity of TGES is tied to the product of the total mass of bricks and the height of the tower. The total mass of bricks is, in turn, associated with the number of bricks, namely, the height of the tower. Consequently, the storage capacity is directly proportional to the square of the tower height [25]. As the height increases, both the system cost and the technical demands for high-altitude operations also rise. Achieving a balance between the two factors requires comprehensive consideration. The tower height is generally limited to within 120m [26]. The specifications of a single brick depend on the technical requirements of the robotic arm and the economic and technical indicators of the entire energy storage system. As the specifications and quality of the single brick improve, the total cost decreases, although the technical

requirements for the robotic arm increase. The enhancement of the technical requirements for the robotic arm is accompanied by increased costs and technical barriers. The ultimate goal should be to achieve simple operation, low cost, and meet energy storage requirements.

The output power (P) of the TGES is calculated by (A2) [25]. The discharge power is influenced by the mass and the speed of descent or ascent of the bricks. The rate of descent or ascent can be adjusted based on output requirements, and increasing the mass or speed is advantageous for enhancing the system's power capacity. However, an increase in the mass of the bricks can lead to a rise in the capital expenditure of the crane [25]. Adjusting the speed is the most cost-effective method for increasing power, and the descent or ascent speed is also connected to the height of the tower. The maximum descent or ascent speed determines the power limit, which is approximately proportional to the height as in (1). It is essential to meet both power requirements and height standards.

$$v_{max} = h/t \quad (1)$$

2.2 Application Practice

In 2019, Energy Vault, a Swiss company [26], deployed an energy storage tower system (outlined in Table 1). The tower, with a height of up to 120 m, features a central tower body equipped with six lifting arms capable of handling concrete bricks weighing up to 35t. These bricks are stacked and dismantled to create the energy storage tower.

The tower's theoretical storage capacity is 35 MWh, utilizing gravity potential energy from the high-speed falling of concrete blocks for rapid and continuous power generation. It achieves a maximum output power of 4 MW within 2.9s, meeting high-speed response demands of the power grid. The energy conversion efficiency is up to 90%, and the locally sourced concrete blocks can be adapted to different regions. The electricity cost is \$ 44.58/MWh.

In 2023, Energy Vault collaborated with China Tianying Company to construct a project in Zhejiang with a storage capacity of 100MWh and an output of 25MW [26].

Currently, a major improvement direction for Energy Vault is to overcome external factors, such as reducing errors caused by weather conditions in the robotic arm, through algorithms.

Table 1. Energy storage parameters of TGES project by Energy Vault [26]

Storage capacity (MWh)	Power output (MW)	Round-trip efficiency (%)	Slope ($^{\circ}$)	Brick quality (t)	Relative height difference (m)
20~100	4~25	80~85	90	35	120

In addition, there are many technologies to improve tower energy storage systems. In 2017, Tan et al. [27] proposed an efficient gravity energy storage (GES) device shown in Fig. 2(a), using movable pulley blocks to lift heavy objects, which effectively reduces energy loss. The comprehensive energy conversion efficiency of the proposed device can reach more than 96%.

To eliminate interference from the external environment on the operation of the energy storage tower, Li et al. [28] proposed a simple single TGES system as in Fig. 2 (b) in 2019, which has the advantage of effectively reducing the impact of strong winds and other climates. Zheng et al. [29] designed a vertical energy storage model as shown in Fig. 2 (c), which uses multiple load-bearing walls to block external interference and forms a lifting channel between load-bearing walls to achieve energy storage and release. Liu et al. [30] proposed a vertical gravity energy storage system in 2021 that utilizes mountain height drop, as shown in Fig. 2 (d). The system collects and stores heavy objects from the top of the mountain, and then causes them to fall along the falling tube. The three designs in Fig. 2 (b) ~ (d) effectively reduce external interference, but their energy storage density decreases and their cost and construction difficulty are both significant, resulting in higher costs compared to other TGES systems.

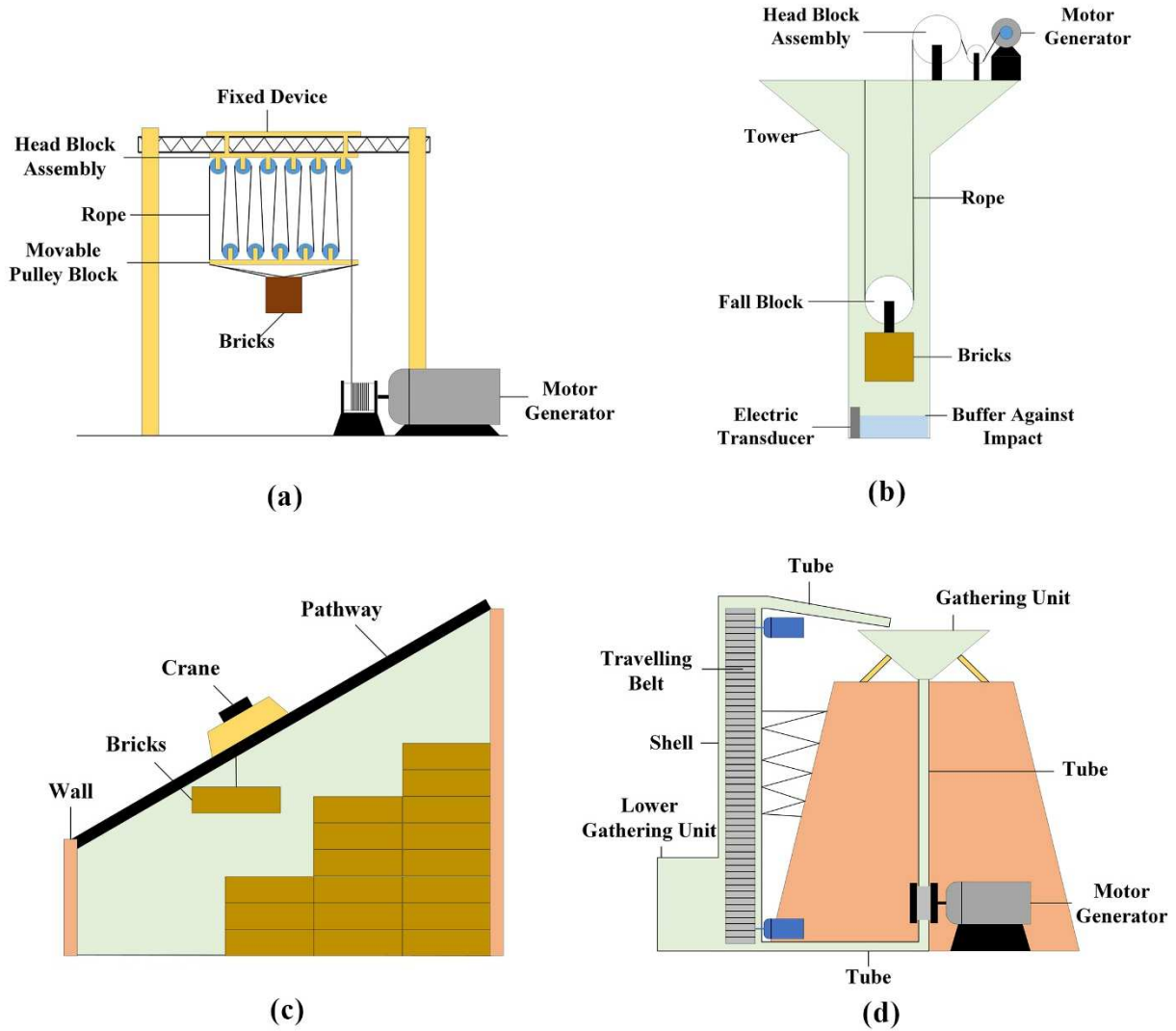


Fig.2 Improving forms of the TGES systems by (a) Tan et al. in 2017^[27], (b) Li et al. in 2019^[28], (c) Zheng et al. in 2021^[29], and (d) Liu et al. in 2021^[30]

2.3 Potential analysis

The tower's height, constrained by the cantilever, limits stored energy, making it suitable mainly for daily and weekly cycles. Its substantial power output, driven by numerous cantilever arms and high-quality bricks, allows rapid energy release within hours for daily cycles or gradual release over days for weekly cycles, addressing challenges posed by integrating diverse renewable energy sources into the power grid.

In a practical example, a 60MW wind power station in Northwest China mandates a 5% energy storage project (equivalent to 6MW) based on regional

directives. Calculating with a rising or falling speed of 2m/s and an 80% round-trip efficiency, the total brick mass required is 191.33 tons.

TGES, being pollution-free with a modular and flexible system, occupies less space vertically compared to pumped storage and lithium-ion battery solutions. Its low site selection requirements make it suitable for most areas. However, the intensive crane boom operations pose challenges for tower crane structures, especially in high-altitude environments. Overcoming these challenges, including external environmental impacts and safety concerns, is a key technical focus for TGES.

In conclusion, the initial consideration should be the impact of tower height on storage capacity and power. Economic analysis should determine the most suitable energy storage height after identifying energy storage demands. Noteworthy are the technical challenges in high-altitude crane operations, highlighting the need for precision and addressing environmental oscillations. Additionally, the integration of vertical TGES with renewable energy, particularly photovoltaic power generation, should be prioritized. TGES proves suitable in areas with significant photovoltaic potential but limited wind energy potential, especially in regions prone to extreme weather events.

3. Mountain Gravity Energy Storage (MGES) and Advanced Rail Energy Storage (ARES)

Since both MGES and ARES can effectively utilize mountain resources and share certain structural similarities, this paper will discuss them together.

3.1. Technical introduction and energy storage/release calculation

3.1.1 Mountain Gravity Energy Storage (MGES)

MGES utilizes the height of the mountain to achieve the mutual conversion and storage of gravitational potential energy and electrical energy [32], enabling

better utilization of mountain resources. The schematic diagram is shown in Fig. 3. An electric motor and generator device are established at the top of the mountain, employing a cable car to transport heavy objects from the bottom of the mountain to the top for energy storage. Conversely, under the influence of gravity, heavy objects descend along the cable in the cable car, generating electricity through the generator. Compared to tower-type energy storage, MGES offers more controllable charging and discharging processes, making it a safer option.

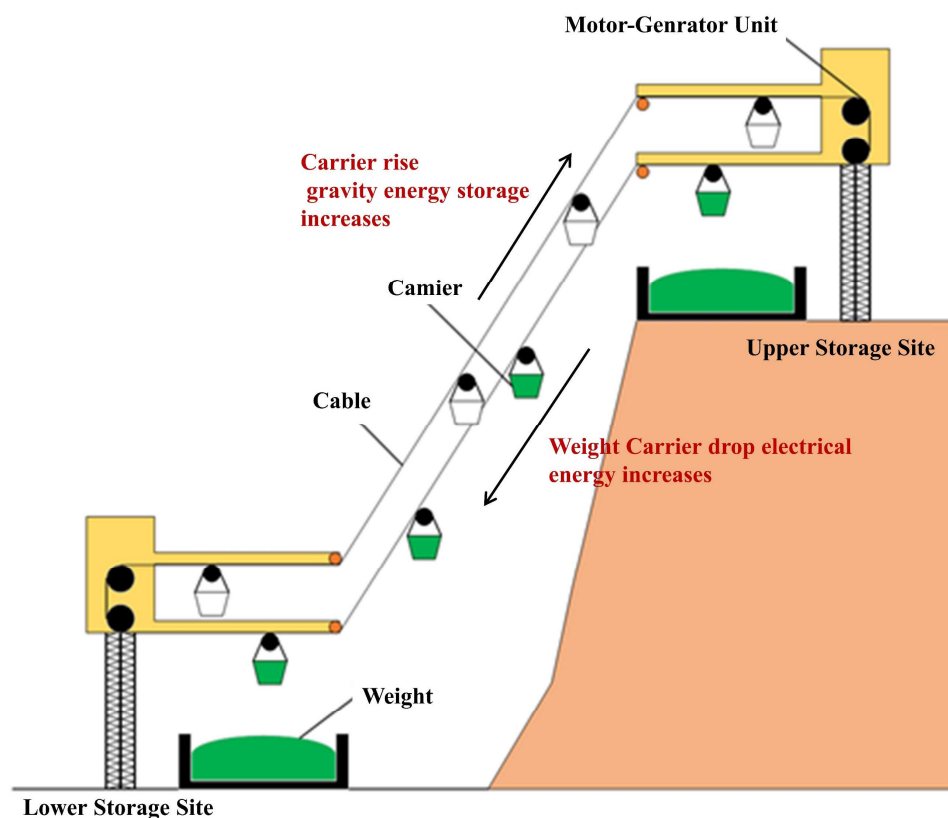


Fig. 3 The schematic diagram of MGES device

The energy storage capacity (E) of MGES device in Fig.4 is calculated by (A3)[33]. Storage efficiency is primarily determined by the product of head loss and motor efficiency. Motor efficiency is determined by the installed equipment, and the impact of head loss on storage efficiency varies based on the storage location and height difference [33]. The relative height difference in existing projects is typically around 200m to 2000m, resulting in head losses ranging from 90% to 97%. With motor efficiency ranging from 75% to 95%, the storage

efficiency is approximately 68% to 92%. Assuming the efficiency of storing and releasing electricity remains consistent, the round-trip efficiency is approximately 46% to 85%. To achieve economic benefits and competitiveness, it is necessary to ensure a round-trip efficiency of over 75%, requiring a relatively high height difference and high motor efficiency.

The output power (P) of MGES is calculated by (A4). The power is related to the time spent. The time spent is negatively correlated with the speed. When it approaches free fall speed (approximately 33m/s), the system's efficiency is close to zero. Therefore, in MGES, the speed should be kept below 10m/s to achieve relatively high efficiency [33]. Speed and slope are interconnected, so mountains with excessive slopes should be avoided to prevent the cable car's speed from becoming too high due to gravity. Electric motors can be employed to control the cable car's descent speed in such cases.

3.1.2 Advanced Rail Energy Storage (ARES)

In addition to cable cars, using railway tracks is another form of GES. The American Railway Energy Storage Company (ARES company, distinguished from ARES) proposed a railway track type for the ARES model in 2014 [34]. The model primarily utilizes existing railway track technology to construct an extensive railway track network, connecting two storage yards with different terrains. When power is needed, the shuttle car, carrying concrete modules, moves from the high storage yard to the low storage yard, and the generator generates electricity integrated into the grid [35]. The schematic diagram is shown in Fig. 4.

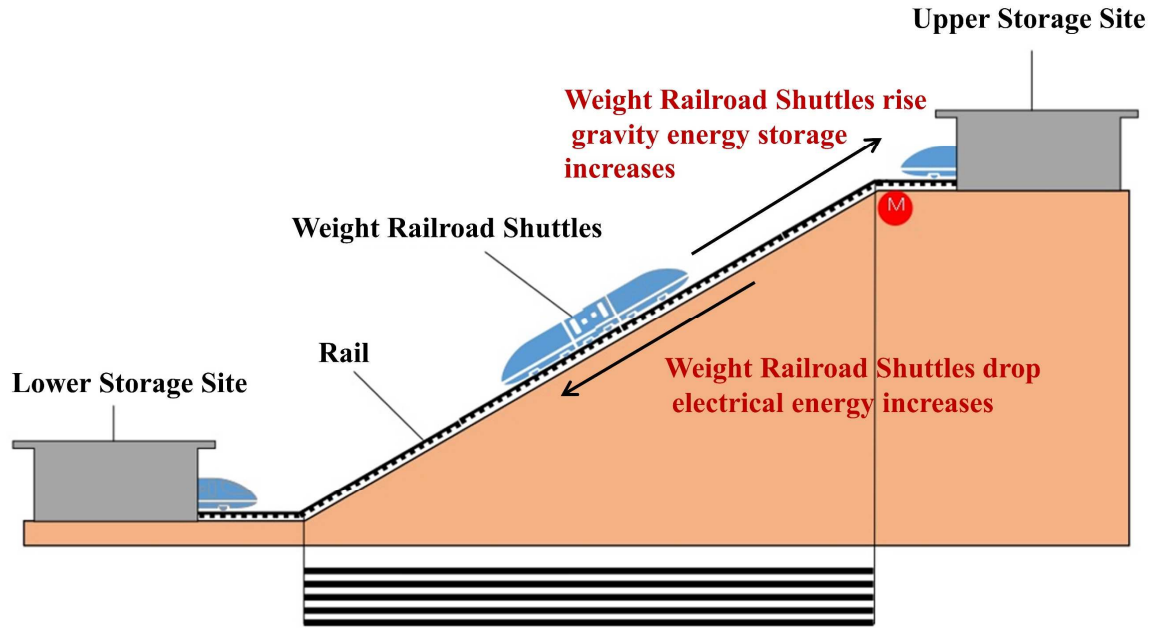


Fig. 4 The schematic diagram of ARES device

The energy storage (E) of ARES device is determined by (A5). The capacity of ARES heavily depends on the construction of the rail network. To ensure the stability of the shuttle vehicle, the slope of the track should not be too steep, and the relative height difference is determined by the capacity requirements. Therefore, the length of the track is typically long. The quality and relative height difference of trains can form a function, and their values can be determined through economic analysis and capacity planning.

The output power (P) of ARES can be calculated by (A6), the speed of the shuttle train v in (A6) can be calculated by (A7). It can be seen from (A7) that the speed is mainly related to the slope and friction coefficient, and the friction coefficient mainly depends on the material and design of the rail.

3.2 Application Practice

In 2012, Peitzke and Brown [36] designed a model consisting of three parts: multiple tracks going up and down, rail cars, and at least one shuttle train track with a transmission and generator. The rate of releasing electrical energy is proportional to the number of shuttle train tracks. Similarly, Tianjin University [37]

made public the use of rails and high-altitude stacking platforms to achieve a low-cost and small-footprint mountain slide model in 2014. The Institute of Electrical Engineering of the Chinese Academy of Sciences [38] proposed a vehicle climbing energy storage system in 2017.

Wei [39] proposed a U-shaped orbital GES device in 2019, which is attached to the slope of a valley to set up a track. In 2021, Beijing Qinghang Science and Technology Co., Ltd. [40] also proposed a train energy storage system, which can realize the peak valley regulation function by running the train carriage with heavy objects on the slope track. The comprehensive efficiency can reach more than 80%, and the storage time is long.

To address the shortcomings of slow conversion efficiency and small capacity, ARES company proposed a model consisting of a large number of rails laid in two railway yards with different heights in 2014 [34]. Railroad speeders slide up and down on the rails to realize energy storage and release. In 2020, construction began in Palermo, Nevada, and will be connected to the power grid. The project will employ a fleet of 210 large cars with a total weight of 75,000 tons, running on a group of 10 closed tracks with a length of 9300m and a slope of 7%, and its energy storage parameters are shown in Table 2 [34]. The highest charging efficiency of this energy storage system has reached 86%, and it can discharge continuously for 8 hours. Compared with the model designed by Peitzke and Brown [36], it has increased the upper limit of the electric energy conversion rate.

In the same year, Luo et al. [41] also proposed to build multiple tracks on mountain slopes and use advanced electric vehicles to store and release the gravitational potential energy of heavy objects. However, the models above have higher costs and slightly lower energy storage density.

Table 2. Energy storage parameters of ARES project by ARES company [34]

Storage capacity (MWh)	Power output (MW)	Round-trip efficiency (%)	Slope (°)	Shuttle train mass (t)	Relative height difference (m)
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100~240,000	5~10,000	78~80	2~5	1,223	610
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The construction of railway tracks increases costs, so the Austrian International Research Institute (IIASA) proposed a MGES model in 2019 [33], with specific parameters shown in Table 3. In comparison to the rails used in the above devices, the MGES model alters the heavy object transportation path and utilizes cables instead of rails to decrease the transportation cost of heavy objects and increase energy storage density. This form of energy storage originates from the American Energy Cache company, which completed the construction of the first engineering prototype in California in 2012.

Table 3. Energy storage parameters of the MGES project by MGES [33]

Storage capacity (MWh)	Power output (MW)	Round-trip efficiency (%)	Slope (°)	Cable car load (t)	Speed (m/s)	Relative height difference (m)	Electric efficiency (%)
2,000~20,000	0.5-20	77~82	>30	1~2	<10	200~2,000	75~95

GES is a viable option for small power grids. This model involves constructing platforms on steep canyons or cliff edges, making it suitable for use in remote areas or on islands. Loading sand and gravel into the cable car is facilitated through an underground gas station, where valves release the sand and gravel stored in the upper and lower storage locations. The energy storage capacity of the MGES system is constrained by the terrain, and the larger the height difference, the more energy it can store. Due to the varying obstruction intensity of the terrain, the cost per MW also varies from \$50 to \$100 (the steeper the mountain, the lower the cost) [42].

In 2020, C-EPRI Electric Power Engineering Co., Ltd. proposed an improvement plan for MGES [43], which uses a motor to drive the conveyor chain, reducing the intermediate links of energy conversion, further reducing costs, and improving the conversion efficiency of the system.

3.3 Potential analysis

When constructing an MGES project, the approximate storage capacity should be determined based on storage needs and benefits. The energy storage medium should be selected from surrounding low-priced or abandoned ores. According to (5), the capacity is related to the total mass and height difference of the ore. Based on the comparison of economic conditions and technical difficulties, the relationship between the total mass and height difference of the ore should be further determined. When the height difference increases, the storage capacity increases, and the head loss decreases, but the cost and technical difficulty correspondingly increase.

Compared to MGES, ARES has more advantages in capacity, while the latter has an advantage in automation work and construction costs. The former can be used for large slopes with good terrain and is dependent on terrain, while the latter is more suitable for remote areas or islands. The potential of GES in mountainous areas of China is enormous, with many suitable sites. Both can enhance people's future utilization of mountain resources and play a significant role in peak shaving and valley filling of mountain power grids.

In the future, ARES should consider how to further increase capacity and achieve better fully automated work, such as expanding the scale of automatic concrete modules for up and down loading and unloading. The main contradiction lies in the low conversion efficiency of short-term devices due to limited operating paths, which cannot meet the demand of the power grid, and the increase in construction costs due to multiple operating paths.

There are many similarities in the site selection and operating principles between ARES and MGES, but there are also several differences that need to be noted:

The first difference lies in the cycle time of the two systems. ARES relies on the robustness of railway tracks and aims to replace pumped storage to some extent. Consequently, the storage platform of ARES has a large capacity, with ARES suggesting that its suitable storage capacity ranges from megawatts to

gigawatts. However, due to ARES's numerous car ferries and high carrying capacity, its power output is relatively large. As a result, it primarily operates in cycles of hours and days to capitalize on the price differences in various periods. ARES claims that its suitable energy storage cycle is 15 minutes to 24 hours and beyond [34]. To achieve a long energy storage cycle, it needs to continuously move the car ferries up and down, resulting in significant wear and tear on the vehicles and smaller benefits compared to short-term energy storage scenarios. On the other hand, MGES is primarily suitable for small islands and other locations with modest energy demands. Its design aims to address the shortfall in pumped storage for energy output below 20MW. However, since its energy storage medium is readily available and storable, and its storage platform exhibits good flexibility, Hunt D and others suggest that its energy storage cycle can extend from weeks to years. This feature is advantageous for addressing the intermittency of various renewable energy sources in small islands, where internet accessibility may be challenging [33].

Due to differing instantaneous demand, the output power also varies. A comparison between Table 2 and Table 3 reveals differences in the power output of ARES and MGES. ARES is designed to replace pumped storage with high-capacity energy storage requirements, necessitating the instantaneous generation of large-capacity electricity. As a result, the output power typically exceeds 20MW, providing ample room for expansion. On the other hand, MGES aims to compensate for the shortfall in pumped storage in low-capacity energy storage, catering to situations where small islands have minimal instantaneous output power demands. Therefore, MGES emerges as the optimal choice for long-term energy storage capacity projects below 20MW. Instead of being competitive, these systems are complementary. Combining the strengths of both ARES and MGES can maximize China's mountain energy storage potential.

As mentioned earlier, the energy storage medium for MGES includes gravel, crushed stone, etc. These materials can be directly obtained from nearby

mountainous areas, resulting in lower costs and almost no need for additional processing. On the other hand, the energy storage medium for ARES typically involves shuttle cars carrying concrete modules. This necessitates consideration of the cost associated with concrete modules and shuttle cars, with high manufacturing difficulty and cost. However, it's worth noting that the new mine project by ARES Company uses gravel from the mine directly [34], so whether there is any difference in energy storage media depends on the specific project planning.

Another crucial difference between the two lies in the devices used to transport energy storage media. MGES utilizes cable cars for transportation, while ARES relies on rail transport. The latter option is more expensive but can be integrated with existing railway systems. Furthermore, railway technology is currently relatively mature and can effectively reduce costs. Given the widespread presence of laid railway tracks globally, some regions may even transform abandoned railway tracks into rail-type gravity energy storage projects. In areas with lower traffic on railway sections, consideration of the possibility of combining with rail-type gravity energy storage projects may be explored.

4. Shaft Gravity Energy Storage (SGES)

4.1 Technical introduction and energy storage/release calculation

The key to storing gravitational potential energy is the creation of height differences. To mitigate challenges related to high-altitude work and minimize interference from the external environment, researchers have proposed developing gravitational potential energy underground, termed as SGES. This energy storage method is safer, and the utilization of abandoned shafts can also contribute to a reduction in construction costs. The specific schematic diagram is shown in Fig. 5.

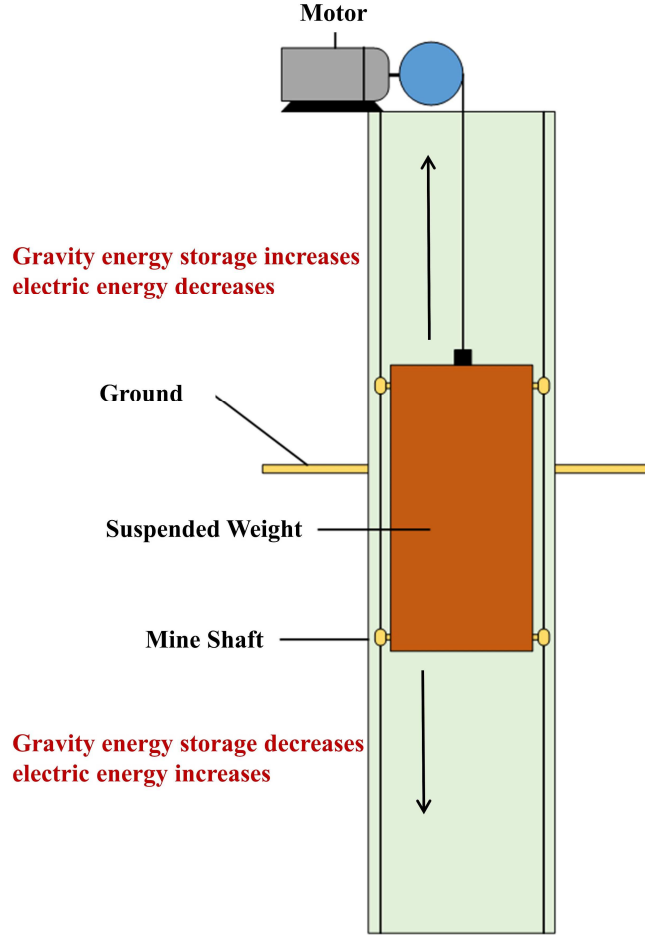


Fig. 5 Schematic diagram of typical SGES device

The energy storage capacity (E) of a SGES device in Fig.5 is calculated by (A8), The mass of suspended heavy objects is determined by its density and volume. Those material with high density but low price is preferred. For a well with circular cross section, the volume of the suspended heavy objects V is determined by the height and diameter of the well, which can be calculated by (2).

$$V = \frac{\pi d^2 h}{4} \quad (2)$$

The selection of the height of suspended heavy objects requires comprehensive consideration of the relationship between the height of the mine and the available height of storage capacity. The available depth D' of storage capacity can be calculated by (3)

$$D' = D - h_s \quad (3) E = \frac{\eta \rho \pi d^2}{4}$$

It can be seen that capacity E is a function of the height of suspended heavy matter. To obtain the maximum capacity, it is necessary to take the derivative of h , as in

$$(5). \quad \frac{\partial E}{\partial h} = \frac{\eta \rho \pi d^2}{4} (D - 2h) \quad (5)$$

Making the derivative equal to 0, the maximum capacity can be obtained when $h=0.5D$. In addition, it is also necessary to consider the maximum load-bearing value of the equipment used for lifting heavy objects by the crane.

Equation (A9) calculates the steady-state output power, and the response time is the duration it takes for the power generated by the motor-operated transfer system to reach the target steady-state power, which is approximately 1 second for SGES. Various output power levels necessitate different motor equipment. The higher the output power, the more stringent the requirements for the motor equipment. Cost-effective motor equipment should be chosen while ensuring the response time.

4.2. Application Practice

Gravitricity, a company based in Scotland, introduced the shaft model [44]. It utilizes giant stones weighing between 500 tons to 5000 tons (with a total capacity of up to 12000 tons) to deliver power ranging from 1 MW to 20 MW within a timeframe of 5 minutes to 8 hours [45]. The system boasts an efficiency level of 80% to 90%. The company demonstrated the use of a cable to lift a 50-ton object and lower it in 11 seconds, allowing the system to reach peak power. In 2021, the company constructed a new 0.25 MW Edinburgh demonstrator and embarked on a 2 MWh energy storage project in Europe in 2022.

SGES exhibits a prolonged lifespan, short response time, and flexibility in choosing operating modes: slow operation at low power consumption or fast operation at high power, all at a significantly lower cost than lithium batteries [46]. However, it encounters various challenges, including the steel cable bundle spreading due to insufficient torque, addressing the shaking and rotation of heavy objects, and addressing the crane's insufficient load-bearing capacity.

Gravity Power Company introduced a GES method in 2011, as illustrated in Fig. 6 (a), which effectively repurposes abandoned mines. The operational process

involves pumping excess electric energy into the deep underground using a water pump. During the lifting of the piston, energy is stored [47].

In many documents, this method is cited as a case. However, in this paper, it is treated as a part of SGES because it utilizes abandoned mines and shares a similar overall framework with other SGES. The primary distinction lies in the transportation mode of the energy storage medium, which is changed from water to a rope. Due to the limited number of actual engineering projects and demonstration projects, it is not discussed separately but is considered as an integral part of SGES. According to Gravity Power, the project aims to return energy to the power grid at a rate of \$37.44/MWh, which is less than half the cost of lithium-ion batteries, inclusive of the energy loss during the round trip, and the project is designed to last more than 40 years.

In 2019, Morstyn et al. also proposed the shaft model [45], as illustrated in Fig. 6 (b), with some parameters presented in Table 4 [44-45]. This model, compared with previous shaft models, is more optimized to achieve fixed weights, thereby reducing excess lateral movement of weights and minimizing collisions. The feasibility of combining SGES with CAES methods was also explored.

Table 4. Energy storage parameters of SGES project by Gravitality company and Morstyn et al. [44-45]

Storage capacity (MWh)	Power output (MW)	Round-trip efficiency (%)	Slope (°)	Well height (m)	Well diameter (m)	Mass of the suspended load (t)	Response time (s)
<25	1~20	80	90	<1,024	<7.55	500~5,000	1

In 2020, utilizing two interconnected abandoned mines, the China University of Mining and Technology constructed multiple channels between the mines at different elevations. This design aimed to facilitate the horizontal transportation and storage of heavy objects and to divide the release of potential energy from heavy objects into a gradient type during operation [48], as depicted in Fig. 6 (c). While this device significantly enhances the capacity of shaft energy storage, it also leads to a certain increase in excess energy consumption. In 2021, the model

proposed by the Institute of Electrical Engineering, Chinese Academy of Sciences, further increases the capacity of shafts and achieves large-scale energy storage by parking more heavy objects on the tracks at the bottom [49].

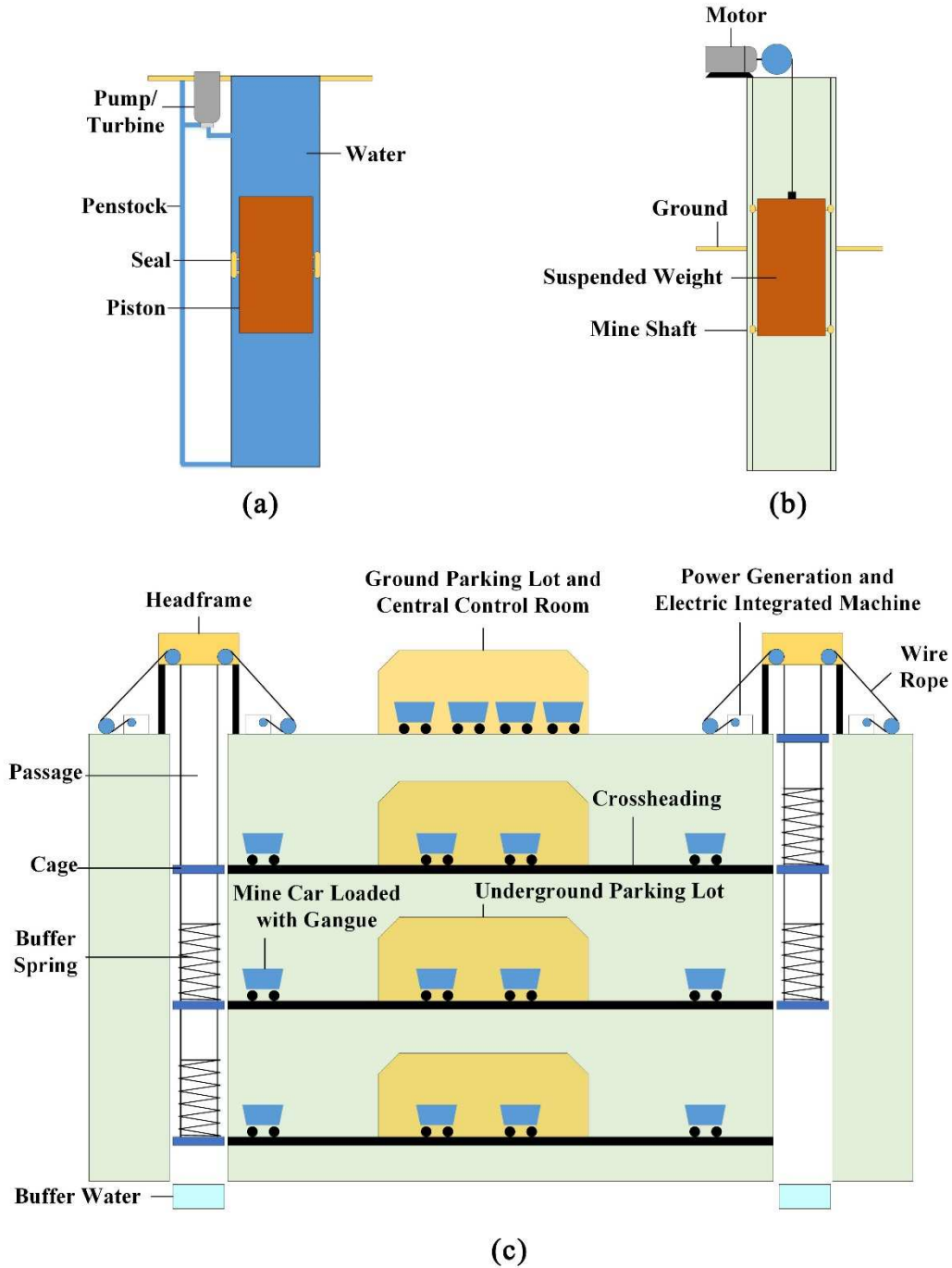


Fig.6 Improving forms of the SGES systems by (a) Gravity Power Company. in 2011^[47], (b) Morstyn et al. in 2019^[45], (c) China University of Mining and Technology. in 2021^[49]

4.3 Potential analysis

Due to the suspension capacity limitation, the energy storage capacity easily reaches its upper limit. Furthermore, future improvements in suspension capacity are constrained by the volume limitations of abandoned mines. Therefore, SGES is suitable for small-capacity energy storage, and suspending heavy objects to a high point for an extended period is prone to consuming more energy, consequently reducing the round-trip efficiency. Thus, SGES is better suited for short energy storage cycles.

Shaft energy storage can be integrated with hydraulic potential energy storage, leading to increased energy storage density compared to standalone hydraulic energy storage. For example, a project proposal by Heindl Energy [50] employs a water pump to hydraulically lift a large rock mass, allowing the rock mass to accumulate potential energy. When the pressurized water is discharged through a turbine, the energy released by the energy storage project is augmented. The project data indicates that the energy storage capacity can range from 1 MWh to 10 MWh, with an investment cost of \$ 120,000/MWh to \$ 380,000/MWh [50]. Additionally, the GES system proposed by Zhang in 2018 combines shaft energy storage with pumped storage [51].

5. Comparison of different gravity energy storage methods

The comparison of theoretical parameters among four types of GES and pumped energy storage is presented in Table 5. In this table, TGES uses Energy Vault's energy storage tower design as an example, MGES takes the model proposed by IIASA as an example, ARES uses the model proposed by ARES company as an example, and SGES takes the model proposed by Gravity Company as an example.

It's important to note that the power cost of different demonstration projects with the same energy storage type can vary. Therefore, the costs shown in the table only have a certain reference value.

Table 5. Comparison of theoretical parameters among four types of GES and pumped energy storage

Methods	Energy density (Wh/m ³)	Power output (MW)	Storage energy (MWh)	Efficiency (%)	LCOE (\$/MWh)	Energy storage circle	Application
PHES	<30	100~2000	1,000~1,000,000	75~80	25.46~171.83	Daily, weekly, monthly, seasonal	Places with large water drop
TGES	>1K	4~20	20~100	90	44.58	Daily, weekly	Flexible location
ARES	>1K	5~10000	100~240,000	75~86	Low cost	Hourly, daily	Flat mountain
MGES	>1K	0.5~20	2,000~20,000	75~82	50.00~100.00	Weekly, onthly, seasonal	Mountains and islands
SGES	>1K	1~20	25	80~90	48.76	Hourly, daily	Abandoned mine

For the four GES forms, the current average energy storage capacity can be arranged from large to small as: ARES>MGES>TGES>SGES.

In terms of power output and energy storage capacity, ARES and MGES exhibit strong flexibility, while TGES and SGES show limited flexibility. The flexibility of energy storage capacity is primarily influenced by the energy storage platform, and the flexibility of output is mainly affected by transportation. The reaction speed is inversely related to the flexibility distribution; TGES and SGES exhibit vertical movement, with the generator located close to the energy storage medium, resulting in a faster reaction speed, while ARES and MGES have slower reactions. From a reliability standpoint, the transportation of ARES and SGES energy storage media is stable, whereas TGES and MGES are significantly affected by wind, leading to poor stability and safety. Regarding the energy storage cycle, MGES is suitable for medium and long-term energy storage cycles spanning weeks to quarters to meet varying energy storage needs. On the other hand, ARES, TGES, and SGES are suitable for short-term energy storage cycles ranging from hours to weeks, facilitating profit generation through price differences in different periods.

Various GES methods encounter challenges. Lifting technology and precise control technology in software algorithms represent key challenges. The former impacts the upper limit of the capacity and storage rate of the GES system, while the latter is indispensable in TGES and SGES. Failure to achieve precise control of heavy objects in specific engineering scenarios can lead to safety issues and compromises in energy storage accuracy during the work process. During the transportation of heavy objects, it is crucial to precisely control the direction of the object, minimize energy consumption as in MGES and ARES, ensure the smoothness of the chain, reduce rail wear, and mitigate safety risks such as shaking of heavy objects and damage to wellbore walls in SGES.

ARES relies on mountainous areas and track laying, MGES depends on mountainous resources, SGES utilizes abandoned mines, and TGES has lower site selection requirements compared to the other three. TGES has the broadest application range, but in regions generating wind power, attention must be given to construction safety due to the impact of wind power on its aerial operations, which cannot be ignored. China, abundant in mountain resources, presents good development prospects for MGES, particularly in small islands and coastal areas. In mountainous regions with suitable track laying and a certain slope, rail-type gravity energy storage exhibits significant development potential and can essentially replace pumped storage. SGES facilitates the reuse of abandoned mines. Each of the four methods has its own advantages and disadvantages and can be applied to various energy storage needs.

The integration of different GES with the electrical grid varies. TGES [26] can effectively provide frequency control services by adjusting the lifting and lowering speed of concrete blocks. It has a significant reserve capacity, enabling a rapid response to additional power needs in the grid. TGES can provide voltage support by adjusting the rate of energy release. It also functions as an energy management tool, storing surplus electricity and releasing stored energy during peak demand periods to balance supply and demand.

MGES, by managing the height difference of cable cars carrying sand and gravel, can swiftly respond to changes in grid demand, supplying additional power or absorbing excess electricity to help maintain grid stability. MGES [33] can dynamically adjust water levels to control the output of generators for frequency control. The system can quickly adapt to fluctuations in grid frequency, providing or absorbing energy to maintain a stable frequency. MGES adjusts the height of cable cars to regulate output power and, consequently, modulate grid voltage. MGES assists in balancing the energy supply and demand in the grid. The system stores potential energy during periods of low power demand and releases stored energy during high-demand periods, effectively managing energy dispatch in the grid.

ARES [53] can provide frequency control services by controlling the speed of train movement, aiding in maintaining grid frequency stability. ARES also serves as part of the reserve capacity, swiftly releasing energy when needed to respond to grid emergencies or sudden demands. Similar to other energy storage technologies, ARES can offer voltage support by adjusting the rate of energy release.

SGES [44] can meet peak demand in the grid by rapidly releasing stored energy, helping balance power supply and demand. SGES can flexibly adjust the operation mode of the energy storage system to provide frequency control and reserve capacity.

6. Conclusion

The rapid development of renewable energy in China is accompanied by a significant amount of waste. From 2015 to 2017, the average light rejection rate in northwest China was 19.57% [52]. In January 2017, the National Energy Administration released "Wind Power Grid-connected Operation in 2016," revealing that the national average "abandoned wind" rate reached 17% [53]. Energy storage can effectively reduce the waste of renewable energy and better

implement the concept of sustainable development. Therefore, the demand for energy storage in the energy industry is increasing, and the outlook for GES is very promising.

Various GES methods have undergone improvements. When constructing GES projects, the most suitable method can be selected through comparative analysis according to local demand, terrain conditions and various factors [54]. Except for ARES, the LCOE costs of the other three energy storage projects are relatively close, all hovering around \$50.00/MWh. Therefore, cost is not a decisive factor in the specific selection process. ARES is suitable for output demands in the range of several thousand MW, while the other three energy storage methods can be chosen for output demands less than a few MW. Additionally, the appropriate energy storage mode can be determined based on factors such as energy storage cycle period and reaction time.

Improving round-trip efficiency is a crucial factor in energy storage engineering, bridging theory to practice. By optimizing the motor output power and transportation path of heavy objects and enhancing the conversion rate of potential energy in heavy objects, including the addition of elastic potential energy (to reduce object kinetic energy consumption), round-trip efficiency can be enhanced. Establishing a physical simulation model for the coordinated operation of GES and the power grid aids in analyzing its dynamic characteristics during grid connection. This, in turn, enhances the understanding of the energy conversion process in energy storage engineering and contributes to further improvements in round-trip efficiency.

The construction of GES projects requires coordination across multiple fields. Attention should be given to the integration with renewable energy during the development process, leading to the emergence of various energy storage methods and combinations. Renewable energy plays a crucial role in sustainable development, making GES closely linked to renewable energy an indispensable component of sustainable development. In the future, within the context of

sustainable development, it is imperative to develop a range of energy storage methods in a coordinated manner, generating electricity collaboratively with various renewable energy sources.

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